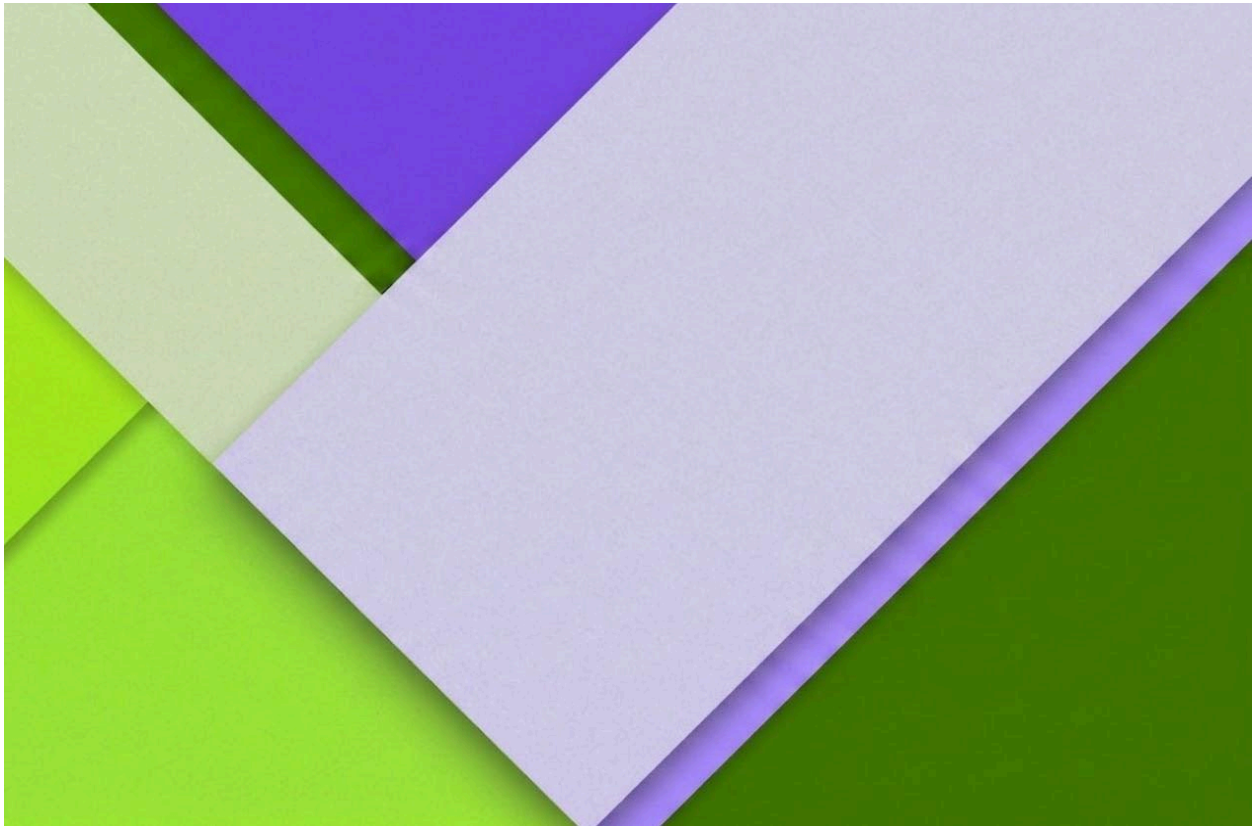




# SID's And NRC

Tuesday, June 02 2026

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## Week 3

Block diagram for Service ID 0x22, 0x2E, 0x2F, 0x31, 0x10, and 0x27  
and NRC (7F) for each

## SIDs

### I. 0x10 - Diagnostic Session Control

None

sequenceDiagram

participant Tester

participant ECU

Tester->>ECU: 10 03

alt Session Supported

ECU-->>Tester: 50 03 00 32 01 F4

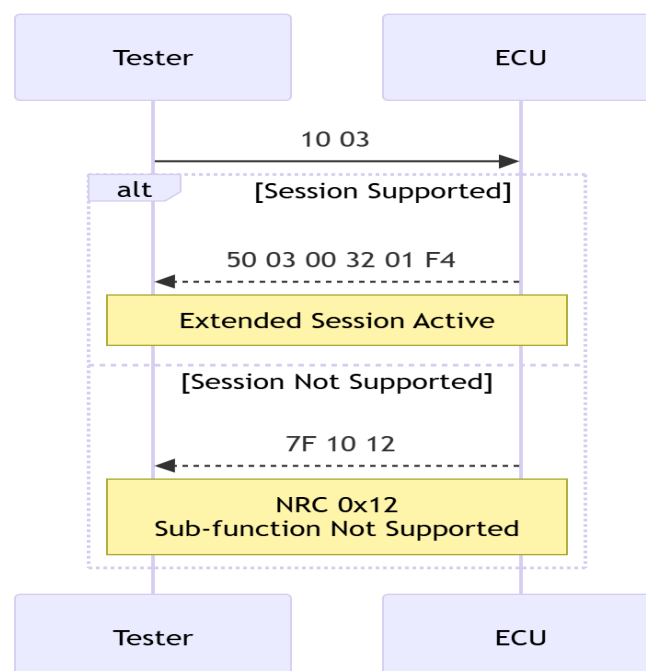
Note over Tester,ECU: Extended Session Active

else Session Not Supported

ECU-->>Tester: 7F 10 12

Note over Tester,ECU: NRC 0x12<br/>Sub-function Not Supported

end



## II. 0x27 - Security Access Control

None

sequenceDiagram

participant Tester

participant ECU

Note over Tester,ECU: Security Access (SID 0x27)

Tester->>ECU: 27 01

Request Seed (Level 1)

ECU-->>Tester: 67 01 Seed

Note over Tester: Calculate Key =  $f(\text{Seed})$

Tester->>ECU: 27 02 Key

Send Key

alt Key Valid

ECU-->>Tester: 67 02

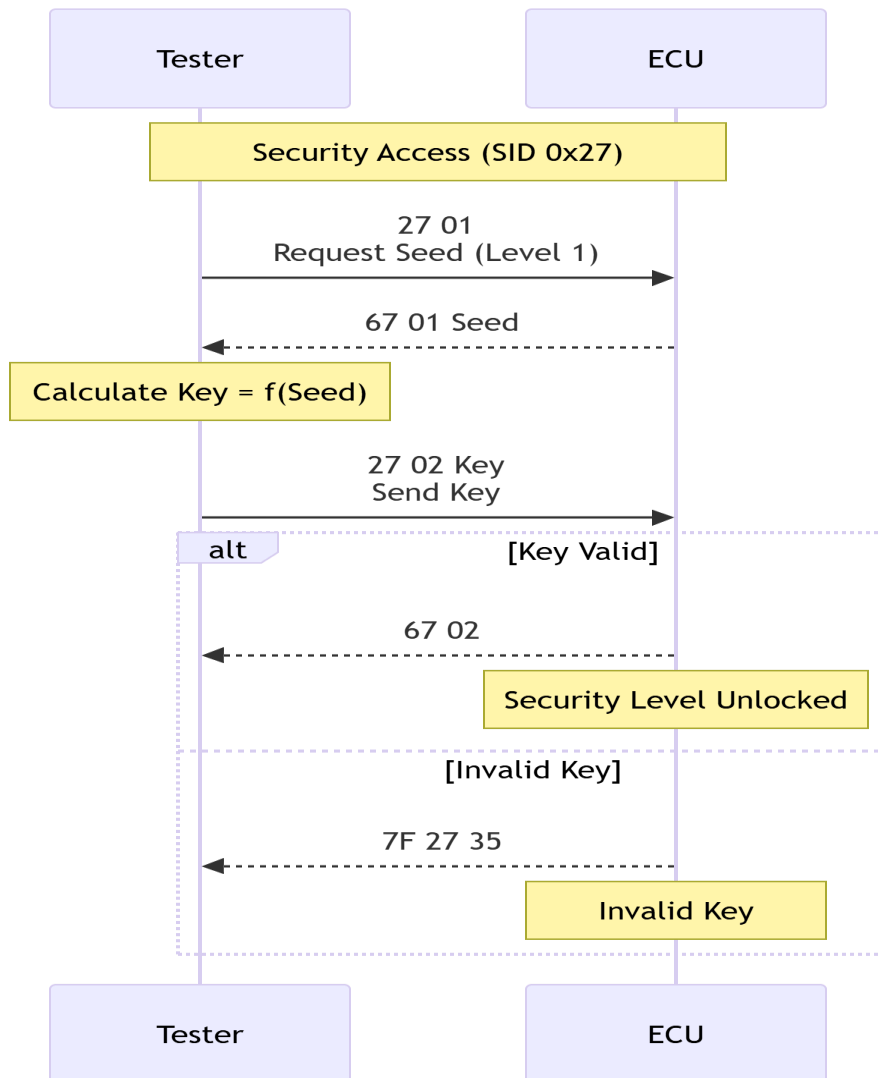
Note over ECU: Security Level Unlocked

else Invalid Key

ECU-->>Tester: 7F 27 35

Note over ECU: Invalid Key

end



### III. 0x22 - Read Data by Identifier

None

sequenceDiagram

participant Tester

participant ECU

Tester->>ECU: 22 F1 90

alt DID Supported

ECU-->>Tester: 62 F1 90 AA BB CC

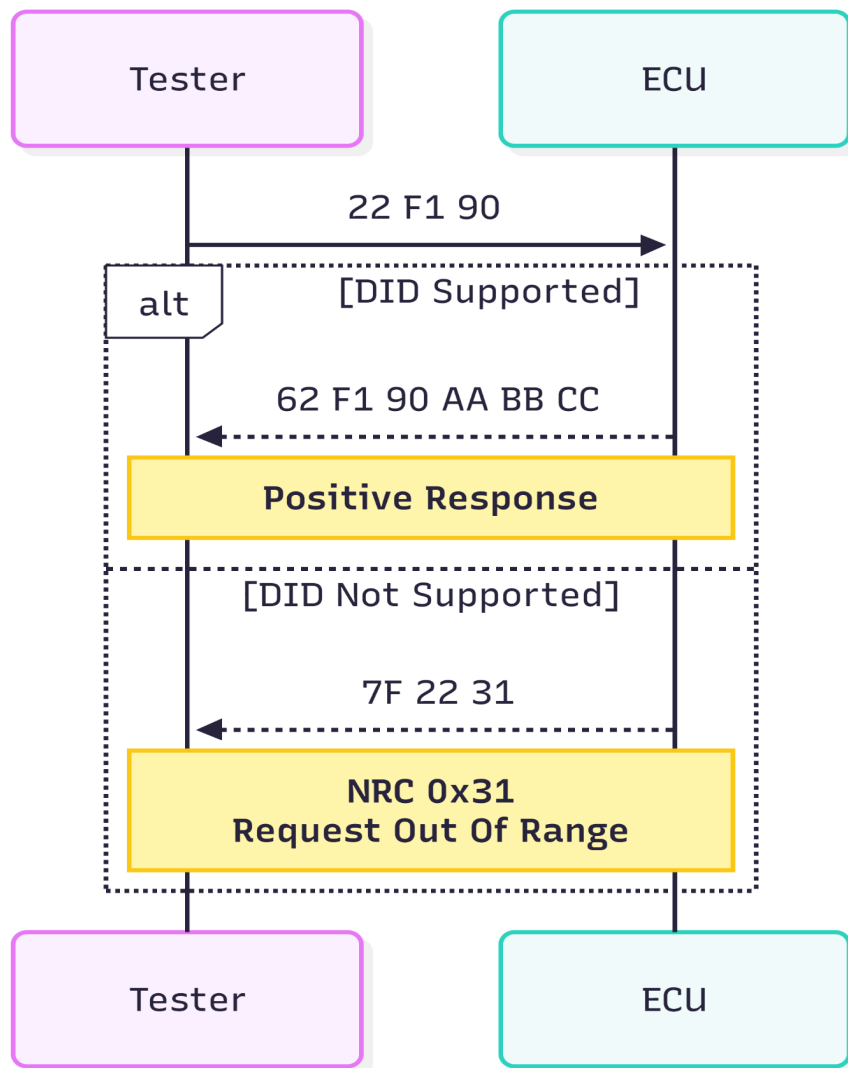
Note over Tester,ECU: Positive Response

else DID Not Supported

ECU-->>Tester: 7F 22 31

Note over Tester,ECU: NRC 0x31<br/>Request Out Of Range

End



## IV. 0x2E - Write Data by Identifier

None

sequenceDiagram

participant Tester

participant ECU

Tester->>ECU: 2E F1 90 AA BB CC

alt Write Successful

ECU-->>Tester: 6E F1 90

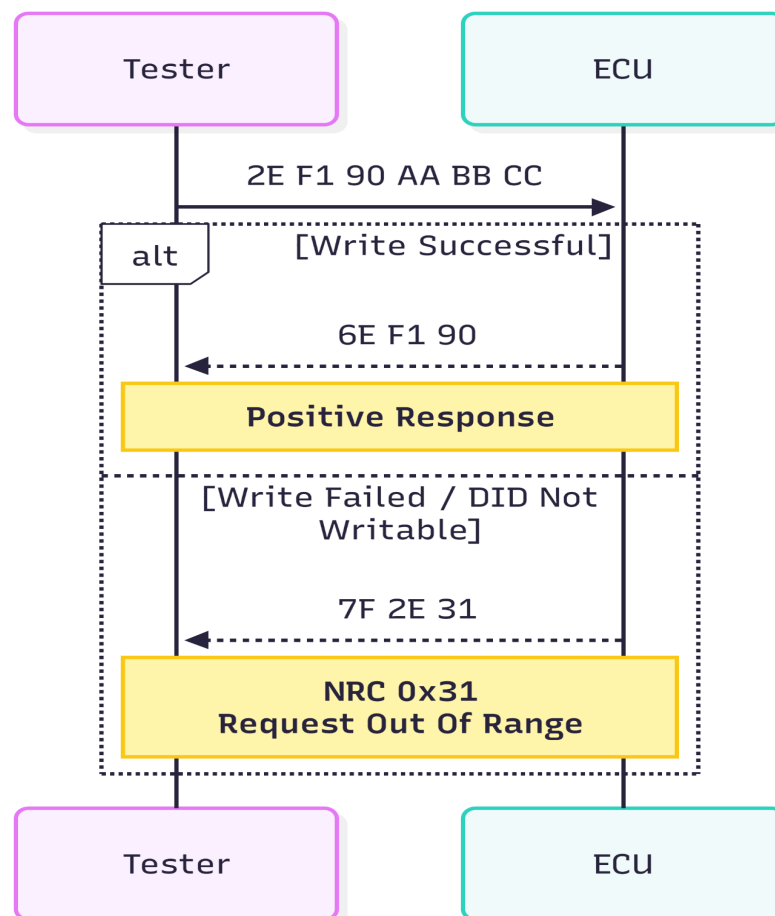
Note over Tester,ECU: Positive Response

else Write Failed / DID Not Writable

ECU-->>Tester: 7F 2E 31

Note over Tester,ECU: NRC 0x31  
Request Out Of Range

end



## V. 0x2F - Input Output Control By Service

Python

sequenceDiagram

participant Client

participant ECU

Client->>ECU: Request: 0x2F [DataIdentifier, ControlOption]

alt Positive Response

ECU->>Client: Response: 0x6F [DataIdentifier, ControlOption]

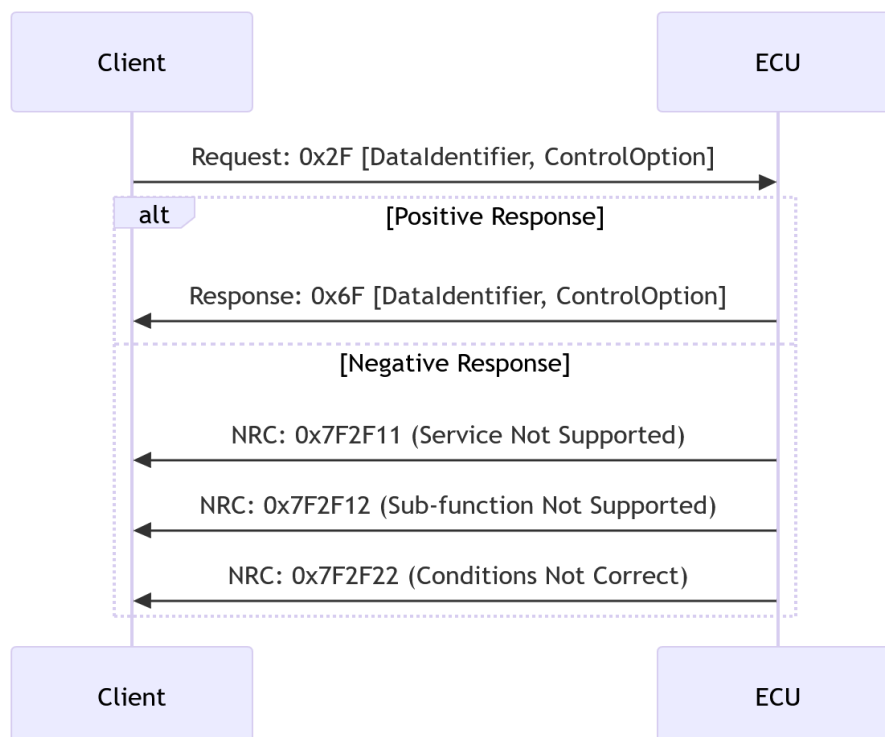
else Negative Response

ECU->>Client: NRC: 0x7F2F11 (Service Not Supported)

ECU->>Client: NRC: 0x7F2F12 (Sub-function Not Supported)

ECU->>Client: NRC: 0x7F2F22 (Conditions Not Correct)

end



## V. 0x31 - Routine Control

None

sequenceDiagram

participant Tester

participant ECU

Tester->>ECU: 31 01 02 03

Note over Tester,ECU: Request Start Routine

alt Routine Started Successfully

ECU-->>Tester: 71 01 02 03

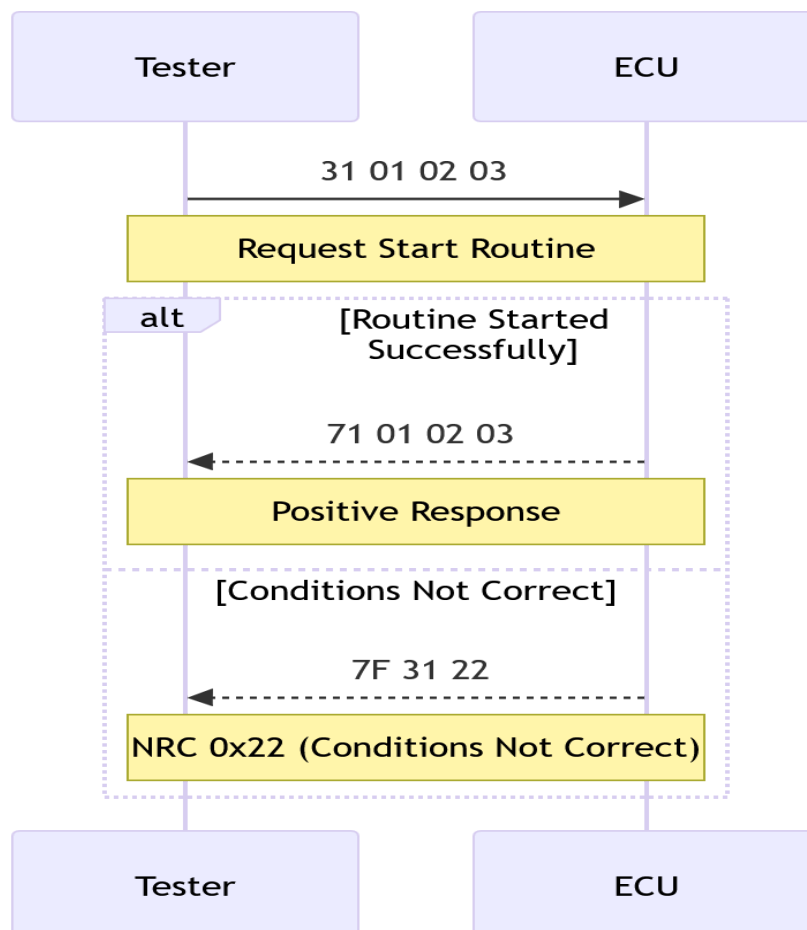
Note over Tester,ECU: Positive Response

else Conditions Not Correct

ECU-->>Tester: 7F 31 22

Note over Tester,ECU: NRC 0x22 (Conditions Not Correct)

end





## UDS Request Structure

| SID         | Service Name                | Request Structure (Hex Bytes)                                                        | Description                                                                                                               |
|-------------|-----------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>0x10</b> | Diagnostic Session Control  | 10 [Sub-Function]                                                                    | Used to change the diagnostic session (e.g., 10 03 for Extended Session).                                                 |
| <b>0x27</b> | Security Access Control     | 27 [Sub-Function]<br>[Key Data (optional)]                                           | Used for authentication. Sub-Function specifies Request Seed (e.g., 27 01) or Send Key (e.g., 27 02 [Key]).               |
| <b>0x22</b> | Read Data by Identifier     | 22 [DID High Byte]<br>[DID Low Byte]                                                 | Used to request data using a 2-byte Data Identifier (DID).                                                                |
| <b>0x2E</b> | Write Data by Identifier    | 2E [DID High Byte]<br>[DID Low Byte]<br>[Data Record]                                | Used to write data to the ECU. Requires DID and the new Data Record.                                                      |
| <b>0x2F</b> | InputOutputControlByService | 2F [DID High Byte]<br>[DID Low Byte]<br>[Control Option]<br>[Data Record (optional)] | Used to command control of an I/O signal. Includes DID, Control Option (Sub-function), and optional data for the control. |
| <b>0x31</b> | Routine Control             | 31 [Sub-Function]<br>[Routine ID (2 bytes)] [Data Record (optional)]                 | Used to start, stop, or check a routine. Sub-Function specifies action, followed by a 2-byte Routine Identifier.          |

## Consolidated UDS Negative Response Codes (NRCs)

| NRC (Hex) | Name                                       | Meaning/Description                                                                                                                                       |
|-----------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x11      | Service Not Supported                      | The requested service is not supported by the ECU.                                                                                                        |
| 0x12      | Sub-function Not Supported                 | Requested session type, security level, or specific sub-function is not supported.                                                                        |
| 0x13      | Incorrect Message Length or Invalid Format | The request length or format is invalid, such as incorrect payload size for a write request or wrong length for a Read Data By Identifier (0x22) request. |
| 0x14      | Response Too Long                          | The requested data size exceeds the ECU's transport buffer capacity.                                                                                      |
| 0x22      | Conditions Not Correct                     | ECU conditions do not allow the action (e.g., engine running, unsafe state for writing data, or wrong voltage conditions).                                |
| 0x24      | Request Sequence Error                     | A prerequisite step was missed, such as sending a Key before requesting a Seed (0x27).                                                                    |
| 0x31      | Request Out Of Range                       | The requested parameter (session, security level, DID, or data value) is outside the supported range or does not exist.                                   |
| 0x33      | Security Access Denied                     | The requested operation requires an unlocked ECU. Authentication via Service 0x27 is needed.                                                              |
| 0x35      | Invalid Key                                | The calculated key sent to the ECU does not match its expectation.                                                                                        |

|      |                                              |                                                                                                                      |
|------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 0x36 | Exceed Number Of Attempts                    | The maximum number of invalid key attempts has been exceeded.                                                        |
| 0x37 | Required Time Delay Not Expired              | The lockout timer is active, and a waiting period is required before retrying security access.                       |
| 0x72 | General Programming Failure                  | An accepted write request failed to be physically saved to EEPROM or Flash memory due to a hardware or memory fault. |
| 0x78 | Response Pending                             | The ECU needs more time to process the request.                                                                      |
| 0x7E | Sub-function Not Supported In Active Session | The requested session is not available from the current session.                                                     |
| 0x7F | Service Not Supported In Active Session      | Service 0x10 is not supported in the current session.                                                                |